MILITARY SPECIFICATION

INSERT SHELF UNIT, MEDICAL INSTRUMENT AND SUPPLY SET CHEST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a metal insert of shelves, suitable for insertion in the top section of the standard No 6 medical service aluminum field chest

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-200/9	Aluminum Alloy Bar, Rod Shapes, Tube and Wire, Extruded, 6063
QQ-A-250/8	Aluminum Alloy 5053, Plate and Sheet
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-766	Steel Plate, Sheet, and Strip, Corrosion Resisting
TT-E-527	Enamel, Alkyd Lustreless
TT-P-636	Primer Coating, Alkyd, Wood and Ferrous Metal
TT-P-1757	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
TT-T-306	Thinner, Syntnetic Resin, Enamels
PPP-B-636	Boxes, Shipping, Fiberboard

FSC 6545

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Military

MIL-C-5541 Chemical Conversion Coatings on Aluminum and Aluminum Alloys

MIL-A-8625 Anodic Coatings for Aluminum and Aluminum Alloys

MIL-P-15328 Primer (Wash), Pretreatment, Blue (Formula No 117-B for Metals)

MIL-C-16775 Chest, Field, Aluminum, Empty

STANDARDS

Federal

FED-STD-66 Steel, Chemical Composition and Hardenability

FED-STD-151 Metal, Test Methods

FED-STD-595 Color

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking for Shipment and Storage

MS 16686 Medical Services Symbol

DRAWING

Defense Personnel Support Center (DPSC)

10018 Inserts for Dental Equipment Set, Prosthetic, Field

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Component parts. The shelf unit shall consist of two shelf retaining frames, part No 25, four removable shelves, part No 26, eight spacers, part No. 27, eight adjustable screws, part Nos. 28 and 29, four pilasters, part No. 30, sixteen adjustable brackets part No 30A and sixteen spare adjustable brackets, and two chest clamps, part No 52 (See Drawing and Bill of Materials)

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3.2 Materials. Except as specified herein, materials shall conform to the applicable portions of the specifications, standards, and drawing (with Bill of Materials) referenced herein, and shall be capable of meeting the requirements shown therein. In addition, the metals shall conform to the chemical requirements of their applicable specifications when tested as specified in Section 4 When materials to be used are not specifically designated, those used shall be of the highest commercial quality, and suitable for the purpose intended.

3.2.1 Aluminum sheet. Aluminum alloy sheet shall conform to QQ-A-250/8, and the temper shall be as indicated on the Bill of Material. Part Nos 25, 26 and 27 shall be fabricated of this material

3.2.2 Aluminum channel. Aluminum alloy channels, part No. 25, shall be 6063-T5 in accordance with QQ-A-200/9.

3.2.3 Steel,carbon. Steel for straps of chest clamp part No 52 shall be tempered carbon steel, composition No. 1020 or 1025 of FED-STD-66. Carbon steel parts shall be cadmium plated in accordance with QQ-P-416 Plating shall be type I, class 2.

3.2.4 Corrosion resisting sheet and strip. Corrosion resisting sheet and strip shall conform to QQ-S-766, compositions 302 or 304

3.2.5 Primer for aluminum parts Primer shall conform to TT-P-1757 color Y.

3.2.6 Primer for ferrous metal parts. Primer shall conform to TT-P-636, Composition G.

3.2.7 Enamel. Enamel shall conform to TT-E-527 Color of the enamel shall approximate 34087 (lusterless olive drab) as shown in FED-STD-595, except that for marking, as specified in 3 5, color shall approximate 37038 (lusterless black) of the same standard (see 6.3).

3.3 Construction. Construction shall conform in all details to the requirements specified herein and as shown on applicable portions of the Drawing (with Bill of Materials). The construction shall be sound and durable in all respects The assembly shall be welded on jigs capable of maintaining required tolerances on finished frames. The insert shelf unit shall fit into a No. 3 medical field chest conforming to MIL-C-16775 (see 6.4) when tested as specified in 4.3.2. The frames shall not move when tested as specified in 4.3.3.

3.3.1 Welding. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society All welds shall be produced by qualified welding operators with satisfactory

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welding equipment using satisfactory filler metal or electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter, any cleaning method which will remove these substances and leave the surface clean is satisfactory. All weld joints shall blend into the adjacent metal in gradual, smooth curves These welds shall be continuous with no visible voids, pin holes, or cracks. All exposed spot welds shall be smoothed sufficiently so as to match adjacent area. Welds shall not be oxidized.

3.4 Finish.

3.4.1 Aluminum alloy. After welding has been completed, but before final assembly, all accessible aluminum surfaces of the shelf unit shall be finisned with enamel as specified herein. Prior to application of the finish, all surfaces shall be thoroughly cleaned free from dirt, oil, grease, welding flux and loose scale with solvents, detergents, or other processes which have no deleterious effect on the surfaces. After thorough cleaning and degreasing, surfaces to be enameled shall be prepared for topcoating by one of the following methods:

(1) Chemical treatment in accordance with MIL-C-5541
(2) Anodic treatment in accordance with MIL-A-8625.
(3) Priming treatment, using primer specified in 3.2.5. thinned and applied by spray in accordance with the directions specified under "Preparation for Delivery" of TT-P-1757, and air-dried for not less than 30 minutes before top-coating.
(4) Prepriming with a wash-primer coat in accordance with MIL-P-15328, and treated as in (3) above.

Topcoating shall be with one coat of enamel as specified in 3.2.7. The enamel shall be reduced by taking four parts by volume of enamel with one part by volume of thinner in conformance with TT-T-306 The reduced enamel shall be applied by spraying and cured by a heat treatment equivalent to baking at 250° Fahrenheit for not less than 45 minutes The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, cracking, acid, alkali, and solvents.

3.4.2 Ferrous metal Ferrous metal parts shall be finished before final assembly by thorough cleaning and chemical degreasing, followed by one coat of primer (see 3.2.6), and then at least one coat of enamel (see 3.2.7).

3.4.3 Film thickness of paint Contractor shall endeavor to control film thickness so that the thickness of a single dried coat of paint is 0.3 to 0.7 mil for primer and 1.0 to 1.5 mil for enamel

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3.5 Identification markings The following markings in enamel (see 3.2 7) shall be distributed on the top surfaces of the top 1/4 inch by 1 inch by 6-7/8 inch strips of the frame halves of part No 25 as follows·

(a) A medical services symbol conforming to type I of MS16686 The symbol shall be of a suitable size, location optional

(b) Left strip. The following information shall be printed in 1/4 inch high characters

6545-00-531-7388

NAME OF CONTRACTOR CONTRACT NO.

(c) Right strip. The following item identification shall be printed in 1/4 inch high capital letters - 2 lines·

INSERT SHELF UNIT, MEDICAL
INSTRUMENT AND SUPPLY SET CHEST

3.6 Workmanship Workmanship shall be first class throughout Frame and shelves shall be square and free from twist Shelves shall assemble correctly and easily. All burrs shall be removed and all sharp corners broken Completed insert shelf units shall be free from defects which detract from their appearance or may impair their serviceability

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1 1 Records Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during performance of the contract and for a period of three (3) years after delivery of the supplies to which such records relate

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the aluminum sheet and channels, and steels, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock No (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4 2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I and II (see 4.2.1, 4 2.2) and 4.2.3.

4.2.1 Sampling for examination Sampling for examination shall be conducted in accordance with MIL-STD-105 The inspection levels and acceptable quality levels shall be as indicated in table I Unit of product for examination purposes shall be one insert shelf unit

Table I Sampling for Examination

Type of defect	Inspection level	AQL (percent Defective)
Major	II	1.0
Minor A	II	4.0
Minor B	S-1	1 0

4 2.2 Examination.

4.2.2.1 Shelf Unit The insert shelf unit shall be examined for defects including, but not limited to, those listed in table II.

Table II. Classification of Defects

Categories	Defects
Critical	None defined
Major	
101	Unit not complete with all specified component parts.

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Table II. Classification of Defects (cont'd)

Categories	Defects
Major	
102	Any component part not free of a crack, tear, rupture, or deformed surface.
103	Any exposed or accessible edge not free of a burr and sharpness.
104	Any weld not sound and smoothly finished.
105	Finish not smooth, even, and adherent.
106	Unit identification markings not complete and correct.
Minor A	
201	Identification marking not completely legible.
202	Enameled surfaces not free of excessively chipped or scratched areas.
203	Color of enamel not as required
204	Not clean, free of grease, oil, dirt, other foreign matter.
Minor B	
251	Dimensions not within tolerances shown in drawing.
252	Any adjustable bracket that fails to assemble and disassemble from slots in pilaster standard
253	Any slot in one of the four pilaster standards that is not aligned with other slots at the same level.

4.2.2.2 Preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 (defects per hundred units.)

Table III. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified
Quantity	Quantity in interior packages and shipping containers not as specified
Packaging and packing materials	Any component missing or damaged
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application

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4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, Inspection Level S-1, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one shelf unit.

4.2.3.1 Sampling for chemical testing of metals. A sufficient amount of each lot of each metal used in the construction of the cases shall be selected for the chemical analysis as specified in 4.3.1.

4.3 Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

4.3 1 Chemical analysis of metals. The samples selected in accordance with 4.2.3.1 shall be analyzed in conformance with methods 111 or 112 of FED-STD-151. In case of dispute, method 111 shall be used for referee analysis.

4.3.2 Fit test. Each of the welded frames, part No 25, shall be approximately positioned in a No. 3 medical field chest (see 6.4). and top and bottom shelves shall be inserted within 6 inches of chest ends. The frames shall be secured in place with the four locking screws, and the remaining two shelves added.

4.3 3 Security test. The coverless sample chest containing only the welded frames, secured as specified in 4.3.2, shall be placed on the floor, long side vertical, and rotated on its sides through 720° The chest shall then be picked up and suspended horizontally upside down.

5 PREPARATION FOR DELIVERY

5.1 Packaging Packaging shall be level A, B or C as specified (see 6 2)

5.1 1 Unit package.

5 1.1 1 Level A. Each insert shelf unit shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, grade V3c. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure, water-proofing and strapping shall be as specified in the appendix of PPP-B-636.

5.1 1.2 Level B Each insert shelf unit shall be packaged in an exterior container designed for a type 2 load and fabricated of not less than 275 pound fiberboard and constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Box shall be closed as specified in the appendix of PPP-B-636.

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5.1.1.3 Level C. Each insert shelf unit shall be packaged in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to the point of delivery called for in the contract or purchase order.

5.2 Packing. No further packing is required.

5.3 Marking Each unit package shall be marked as required for the exterior container specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The complete insert shelf unit, when used with the 6545-00-914-3510 No. 6 Medical Instrument and Supply Set Field Chest, is intended to contain a portion of the components of either the 6545-00-299-8592 Dental Equipment Set, Prosthetic, Field, Part A, or of the 6545-00-299-8593 Dental Equipment Set, Prosthetic, Field, Part B.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number and Item Identification.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Availability of color chips. Individual chips of the colors specified in 3.2.7 are available from:

Commissioner
Federal Supply Service
General Services Administration
Standardization Division, Paints Branch
Washington, D C 20406

6.4 Sample chest A 6545-00-914-3480, No. 3 Medical Instrument and Supply Set Field Chest may be obtained on a loan basis upon application to the Contracting Officer and payment of the required deposit

6.5 The insert shelf unit covered by this specification is listed in the Federal Supply Catalog as follows:

National Stock Number	Item Identification
6545-00-531-7388	Insert Shelf Unit, Medical Instrument and Supply Set Chest.

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number: 6545-0025

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DOCUMENT IDENTIFIER AND TITLE
MIL-I-37238, Insert Shelf Unit, Medical Instrument and Supply Set Chest

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
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1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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